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## BRITALINE

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**DESCRIPTION:**

**BRITALINE TABLET 2.5 MG:** A white, round tablet, 7 mm in diameter with marking 'duo' and 'd'.

**COMPOSITION:**

**BRITALINE TABLET 2.5 MG:** Each tablet contains Terbutaline Sulphate 2.5 mg

Preservative: Benzalkonium Chloride 0.04 mg/ml

**PHARMACODYNAMICS:** The tertiary butyl group attached to the terminal nitrogen of the terbutaline molecule is thought to confer selective stimulation of the pulmonary beta<sub>2</sub> receptors and only relatively minor stimulation of cardiac beta<sub>1</sub> receptors. The presence of the two phenolic hydroxyl groups in the meta positions confers resistance to metabolism by the enzyme catechol-o-methyl transferase.

**PHARMACOKINETICS:** The potent bronchospasmolytic effect is rapid in onset and reaches a maximum about 30 minutes after subcutaneous injection, 1 hour after aerosol and 2-3 hours after oral administration. The duration of action is between 4 and 5 hours. In addition to its bronchospasmolytic effect, terbutaline has also been shown to improve mucociliary clearance. Metabolism of terbutaline sulphate which is ingested orally or swallowed following aerosol administration is principally by conjugation in the gastrointestinal mucosa. The drug is absorbed unchanged from the respiratory tract and is excreted mainly as such in the urine. Practically all of an administered dose of terbutaline is eliminated after 72 hours.

**INDICATIONS:** For the relief of severe bronchospasm in bronchial asthma and in chronic bronchitis, emphysema and other bronchopulmonary disorders in which bronchospasm is a complicating factor.

**RECOMMENDED DOSAGE:**

Dosage should be individualized.

**Adult: Oral: Tablet (2.5 mg):** 2.5 to 5 mg up to 3 weeks daily (2.5 mg, 3 times daily is recommended during the first 1-2 weeks).

**Impaired hepatic function:** Hepatic failure has not been shown to influence the metabolism of terbutaline. However, caution should be exercised in patients with impaired liver function.

**Impaired renal function:** As terbutaline sulphate is largely excreted in urine, caution should be exercised in patients with renal impairment.

**CONTRAINDICATIONS:** Hypersensitivity to sympathomimetic amines, aerosol propellants or to any other ingredient.

**WARNINGS AND PRECAUTIONS:**

1. **Tocolysis:** Serious adverse reactions including death have been reported after administration of terbutaline/ salbutamol to women in labor. In the mother, these include increased heart rate, transient hyperglycaemia, hypokalaemia, cardiac arrhythmias, pulmonary oedema and myocardial ischaemia. Increased fetal heart rate and neonatal hypoglycaemia may occur as a result of maternal administration.
2. **Cardiovascular diseases:** Caution is advised when terbutaline is administered to patients with hypertension, coronary artery disease, arrhythmias and tachyarrhythmia. In a few patients, therapeutic doses of terbutaline have been reported to cause some ECG changes but the exact significance of these changes is not known.  
Some investigators consider these changes are not drug related. However, the arrhythmogenic potential of beta<sub>2</sub> adrenoceptor stimulants should be considered.
3. **Diabetes:** Due to the blood-glucose increasing effects of beta<sub>2</sub> stimulants, extra blood glucose controls are initially recommended when diabetic patients are commenced on terbutaline.
4. **Sensitivity to sympathomimetic amines:** Some patients may be unusually sensitive to beta-adrenergic stimulants. Terbutaline should be used with caution when an increased susceptibility to sympathomimetic amines can be expected for instance in patients with hyperthyroidism not yet under adequate control.
5. **Nebulising solution / pulmonary conditions:** Positive pressure delivery systems (IPPB) for respiratory drugs should not be used in pulmonary conditions involving pneumothorax, air cyst or mediastinal emphysema unless special drainage is carried out.
6. **Lack of response:** If the usual dose does not provide the usual relief, a non-responsive state may be developing in which case sympathomimetic aerosols may prove to be harmful (e.g. worsening of hypoxaemia with resultant profound physiological disturbances) as well as ineffective. If a previously effective dose lasts less than usual, patients should be instructed to consult a doctor.
7. **Arrhythmogenic potential:** Beta<sub>2</sub> stimulants have an arrhythmogenic potential which must be considered for each patient when receiving treatment for bronchospasm.

Potentially serious hypokalaemia may result from beta<sub>2</sub>-agonist therapy mainly from parenteral and nebulised administration. Particular caution is advised in severe asthma as this effect may be potentiated by concomitant treatment with xanthines derivatives, steroids, diuretics and by hypoxia. It is recommended that serum levels are monitored in such situations.

**Impaired hepatic function:** Hepatic failure has not been shown to influence the metabolism of terbutaline. However, caution should be exercised in patients with impaired liver function.

**Impaired renal function:** As terbutaline sulphate is largely excreted in urine, caution should be exercised in patients with renal impairment.

**INTERACTION WITH OTHER MEDICAMENTS:** Care is recommended if it is proposed to administer terbutaline in concomitant therapy with other sympathomimetic amines as excess sympathetic stimulation may occur. Beta-adrenergic blocking drugs may inhibit the bronchodilating effect of sympathomimetic bronchodilators and may increase airway resistance in asthmatic patients.

Terbutaline sulphate should be administered with caution to patients being treated with monoamine oxidase inhibitors or tricyclic antidepressants, since the action of terbutaline sulphate on the vascular system may be potentiated.

Concurrent use of terbutaline with diuretics, xanthines or high doses of corticosteroids may increase the risk of hypokalaemia. The antihypertensive effects of diuretics may be reduced when these medications are used with terbutaline.

**PREGNANCY AND LACTATION:**

**Use in pregnancy:** Although no adverse effects in pregnant women or their foetuses have been reported, care with terbutaline sulphate, as with all other drugs, is recommended during the first 3 months of pregnancy.

As maternal pulmonary oedema and myocardial ischaemia have been reported during or following premature labour in patients receiving beta<sub>2</sub>-agonists, careful attention should be given to fluid balance and cardio-respiratory function, including ECG monitoring. If signs of pulmonary oedema and myocardial ischaemia develop, discontinuation of treatment should be considered.

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Due to the risk of pulmonary oedema and myocardial ischaemia that has been observed during the use of beta2-agonists in the treatment of premature labour, before these products are given to any patients with known heart disease, an adequate assessment of the patients cardiovascular status should be made by a physician experienced in cardiology.

**Use in lactation:** Although terbutaline is secreted into breast milk, and milk concentrations are approximately those in maternal plasma, two individual case studies indicate that the infant is likely to receive 0.2 - 0.7% of the maternal dose (0.4 and 0.7 mcg/kg/day respectively, depending (for example) on the time of feeding in relation to administration of the drug. In the 4 infants studied this did not result in any signs of beta-adrenoceptor stimulation.

**SIDE EFFECTS:** The incidence and severity of particular side effects depends on the dose and rate of administration. An initial dose titration will often reduce side effects. At recommended therapeutic doses, the frequency of side effects is minimal.

**More Common Reactions:** Commonly observed side effects include tremor, nervousness, increases in heart rate and palpitations.

**Less Common Reactions:**

**Cardiovascular:** Ectopic beats.

**Gastrointestinal:** Nausea, vomiting, bad taste, diarrhoea.

**General:** Sweating

**Electrolytes:** Potentially serious hypokalaemia may result from beta-2 agonist therapy. **Muscular Skeletal:** Muscle twitching, cramps. **Nervous system:**

Drowsiness, headache, dizziness **Dermatological:** Rash

**Serious or Life Threatening Reactions:** Overdose of terbutaline preparations may produce significant tachycardia, arrhythmia and hypotension.

**SYMPTOMS AND TREATMENT OF OVERDOSE:** The median subcutaneous lethal dose of terbutaline sulfate in mature rats is approximately 165 mg/kg (approximately 90 times the maximum recommended daily oral dose for adults on an mg/m<sup>2</sup> basis). The median subcutaneous lethal dose of terbutaline sulfate in young rats is approximately 2000 mg/kg (approximately 1100 times the maximum recommended daily oral dose for adults on an mg/m<sup>2</sup> basis). The expected symptoms with over dosage are those of excessive beta-adrenergic stimulation and/or occurrence or exaggeration of any of the symptoms listed under ADVERSE REACTIONS, eg. seizures, angina, hypertension or hypotension, tachycardia with rates up to 200 beats per minute, arrhythmias, nervousness, headache, tremor, dry mouth, palpitation, nausea, dizziness, fatigue, malaise, and insomnia. Hypokalemia may also occur. Too frequent administration, as with other sympathomimetic agents, may cause nausea, headaches, and changes in blood pressure, anxiety, tension, restlessness, insomnia, tremor and excitement. The symptoms and signs are those characteristic of excessive sympathetic stimulation.

**Treatment:**

- a) **Mild and moderate cases:** Reduce the dose.
- b) **Severe cases:** Determination of acid-base balance, blood sugar and electrolytes. Monitoring of heart rate and rhythm and blood pressure. Metabolic changes should be corrected. A cardioselective beta-blocker (eg. metoprolol) is recommended for the treatment of arrhythmias causing haemodynamic deterioration. The beta-blocker should be used with care because of the possibility of inducing bronchoconstriction. If the beta2-mediated reduction in peripheral vascular resistance significantly contributes to the fall in blood pressure, a volume expander should be given.

**PACKING/PACK SIZES:**

Blister pack of 3 X 10's and 10 X 10's tablets in a box.

**STORAGE CONDITION:** Store below 30°C, protect from light.  
Keep out of reach of children. *Jauhkan daripada kanak-kanak.*

**SHELF LIFE:** Please refer to outer package.

**PRODUCT REGISTRATION HOLDER & MANUFACTURER:**

DUOPHARMA (M) SDN BHD (42491 M)  
Lot 2599 Jalan Seruling 59 Kaw 3  
Taman Klang Jaya, 41200 Klang  
Selangor Darul Ehsan  
MALAYSIA